

Table of Contents

1. Abstract.....	1
2. Introduction	1
3. Objectives:.....	2
4. Geometry Modeling and CAD Design.....	2
5. Finite Element Analysis	4
6. Analysis Procedure.....	6
7. Results and Discussion	7
8. Comparative Analysis	10
9. Conclusions	11

1. Abstract

The primary objective of this Open-Ended Laboratory (OEL) work is to analyze the stress and strain behavior of an automotive screw jack under static loading conditions. The geometric modeling and CAD design of the screw jack were developed using SolidWorks, and the model was imported into ANSYS Workbench via IGES format for static structural analysis. A finite element mesh was generated using appropriate sizing controls, and adaptive meshing was applied with slightly increased relevance to ensure adequate mesh quality and solution accuracy.

Initially, the screw threads were included in the model; however, their inclusion significantly increased the computational time. Since the threads do not substantially influence the overall stress distribution of the assembly, they were neglected to improve computational efficiency without compromising the accuracy of the results.

Two boundary conditions were applied in the analysis: a fixed support at the base plate of the screw jack and a vertical load applied at the saddle in the negative Y-direction. For each saddle height configuration, the applied load was varied from 1000 N to 6000 N, and the corresponding von Mises stress, Y-direction deformation, and factor of safety were evaluated.

The analysis was performed at saddle heights of 4.75 in, 8 in, 12 in, and 15.56 in to investigate the effect of lifting height on the structural response of the screw jack. The obtained results provide insight into the mechanical behavior, load-carrying capacity, and safety of the screw jack at different lifting heights and loading conditions. A comparative discussion is presented to assess the structural performance and safety of the screw jack under various operating configurations.

2. Introduction

Automotive screw jacks are widely used lifting devices designed to raise vehicles safely for maintenance and repair operations. These components are subjected to significant compressive loads and bending effects, especially at higher lifting heights. Understanding the stress distribution and deformation characteristics of a screw jack under static loading is essential for ensuring structural integrity and operational safety.

The objective of this Open-Ended Laboratory (OEL) work is to numerically investigate the stress, deformation, and safety of an automotive screw jack using finite element analysis (FEA). The influence of lifting height and applied load on the mechanical response of the screw jack is systematically studied using ANSYS Workbench.

3. Objectives:

The specific objectives of this OEL work are:

- To develop a 3D CAD model of an automotive screw jack using SolidWorks.
- To perform static structural analysis using ANSYS Workbench.
- To evaluate von Mises stress, deformation, and factor of safety under different loads.
- To study the effect of saddle height on the structural behavior of the screw jack.
- To assess the overall safety and load-carrying capability of the screw jack.

4. Geometry Modeling and CAD Design

The complete geometric model of the automotive screw jack assembly was developed in SolidWorks. The main components included:

- Base plate
- Screw shaft
- Saddle (top plate)
- Supporting arms and housing

Initially, screw threads were modeled to represent realistic geometry. However, due to the high computational cost associated with meshing fine thread details and their negligible influence on global stress distribution, the threads were removed. The simplified geometry maintains the structural stiffness and load path while significantly improving computational efficiency.

The finalized CAD model was exported in IGES format and imported into ANSYS Workbench for further analysis.

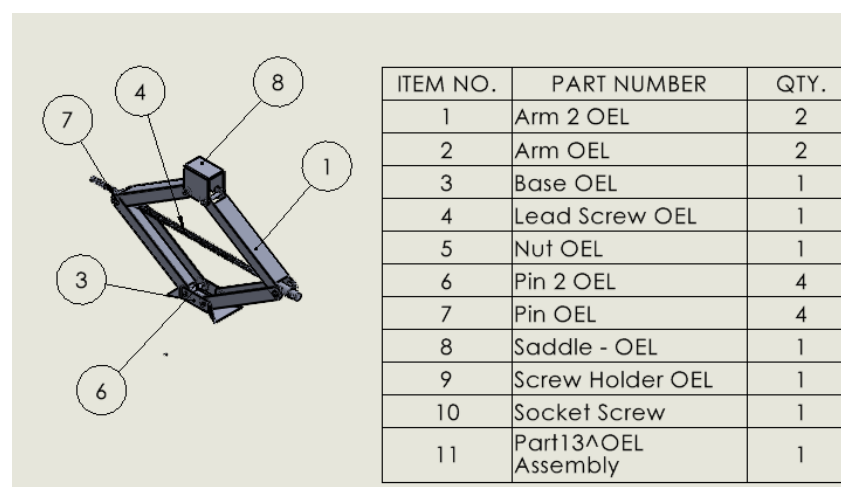


Figure 1: Assembly Details.

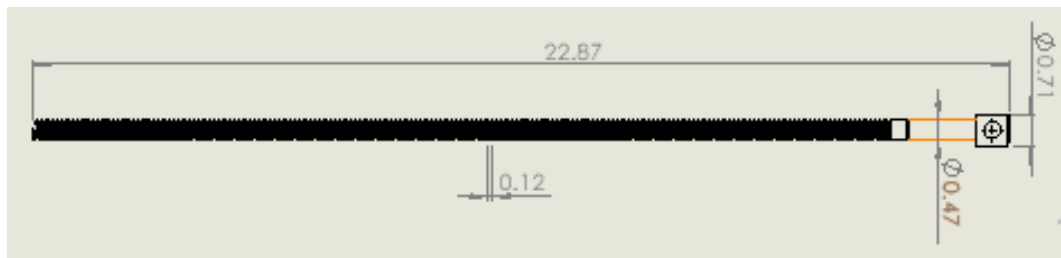


Figure 2:Lead Screw Dimensions.

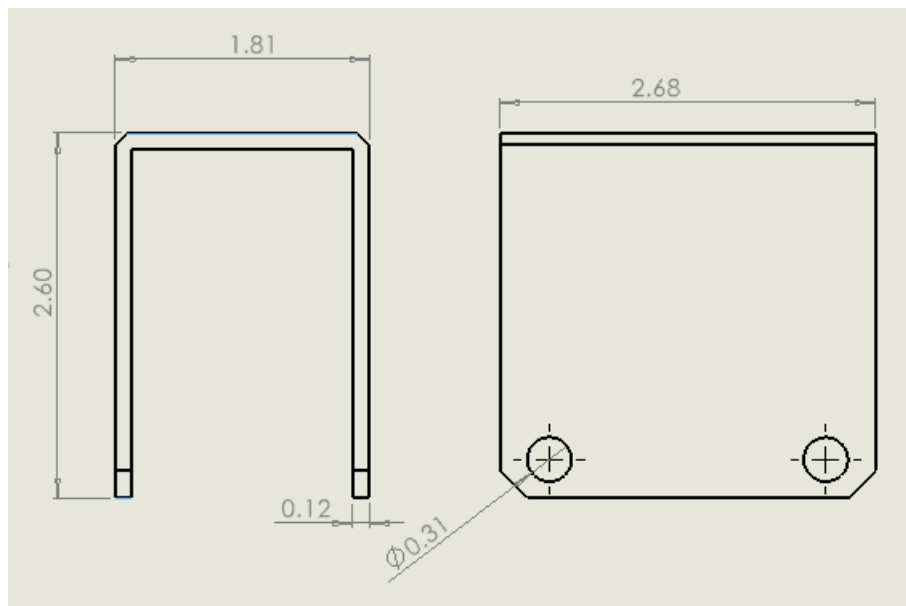


Figure 3:Saddle Dimensions.

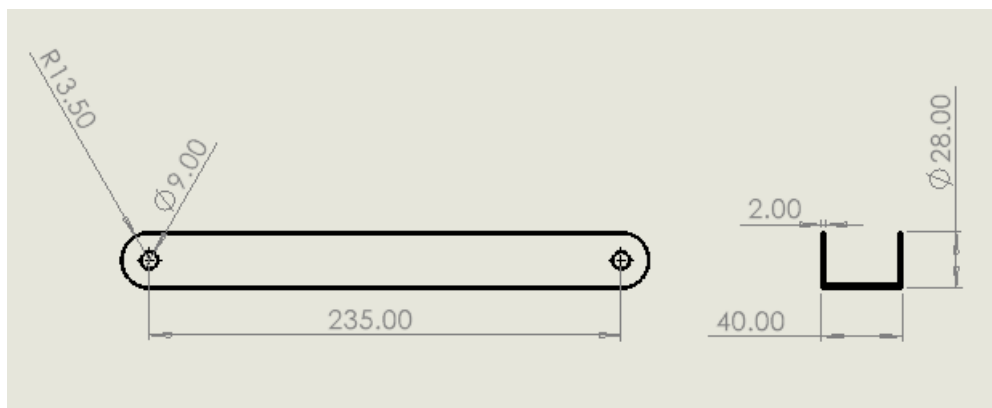


Figure 4:Arm Dimensions.

5. Finite Element Analysis

Material Properties

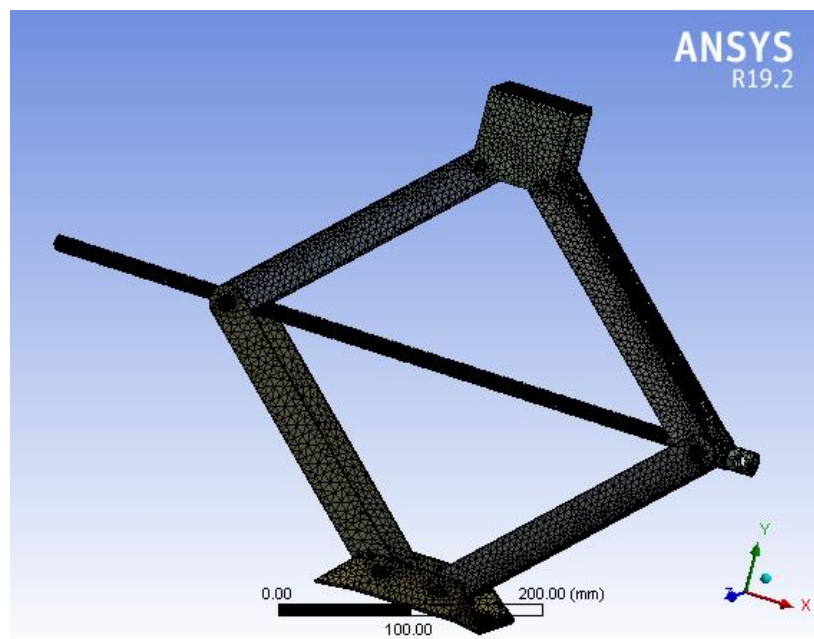
The screw jack components were assumed to be made of structural steel. The material properties used in the analysis are listed below:

- Young's Modulus, $E = 200 \text{ GPa}$
- Poisson's Ratio, $\nu = 0.3$
- Density, $\rho = 7850 \text{ kg/m}^3$

Meshing Strategy

A finite element mesh was generated using ANSYS Mechanical. The following meshing approach was adopted:

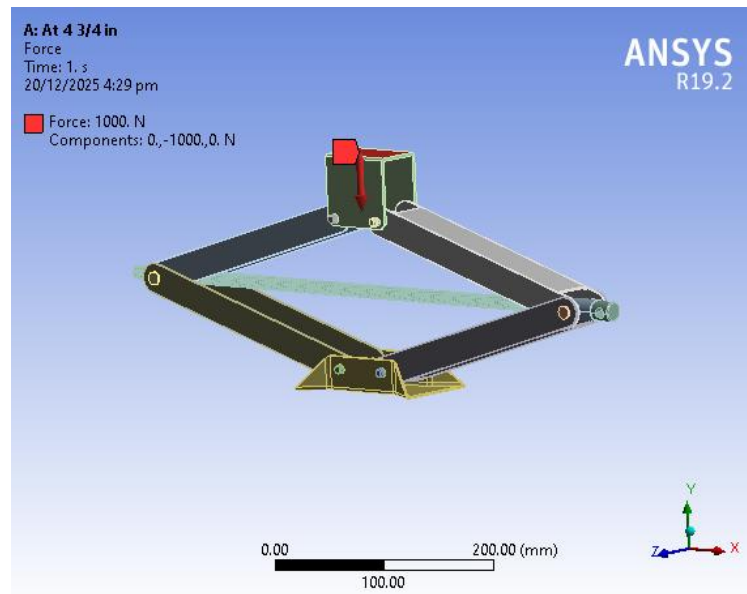
- All Quad elements were used for more accuracy.
- Appropriate sizing controls were applied to critical regions such as the screw shaft and saddle.
- Adaptive meshing with slightly increased relevance was enabled to improve mesh quality.



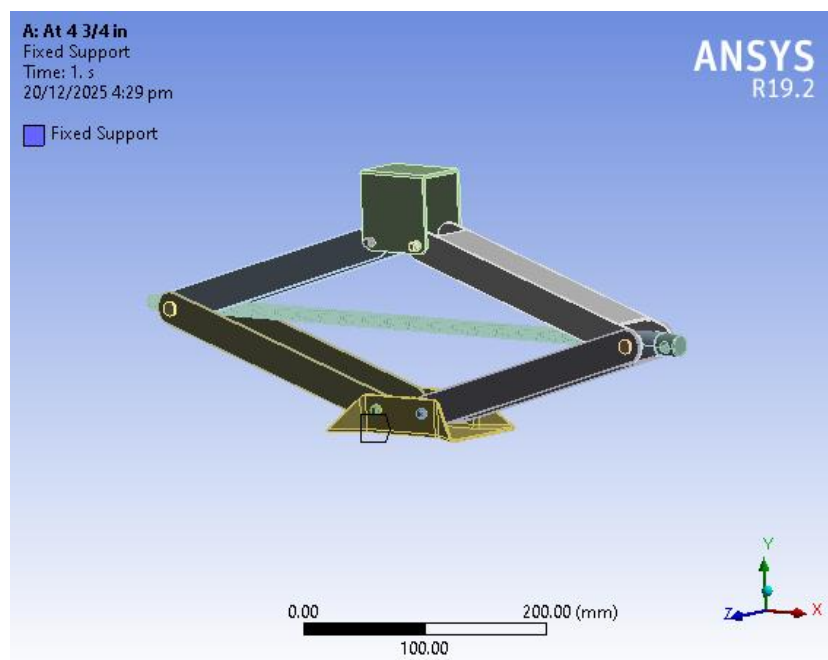
Boundary Conditions

Two primary boundary conditions were applied:

- **Fixed Support:** Applied at the base plate of the screw jack to restrict all translational and rotational degrees of freedom.



- **Load Application:** A vertical compressive load was applied at the saddle in the negative Y-direction.



Loading Conditions

For each saddle height configuration, the applied load was varied as follows:

- 1000 N
- 2000 N
- 3000 N
- 4000 N
- 5000 N
- 6000 N

Saddle Height Configurations

The analysis was carried out for four different lifting heights:

- 4.75 in
- 8 in
- 12 in
- 15.56 in

These configurations represent typical operating positions of the screw jack during use.

6. Analysis Procedure

The static structural analysis was performed using the following sequential steps:

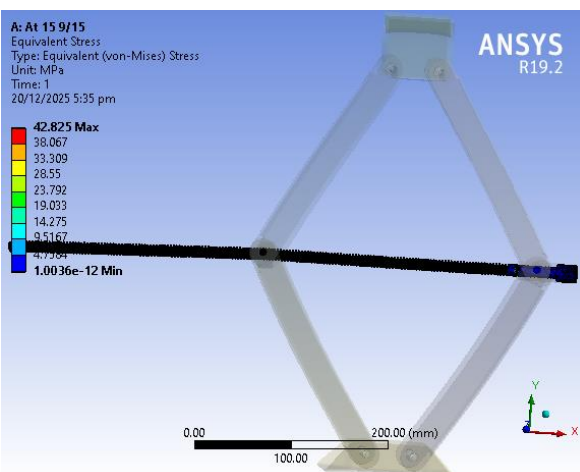
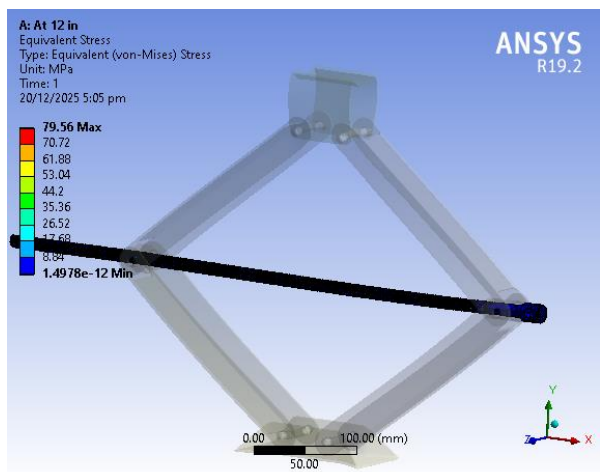
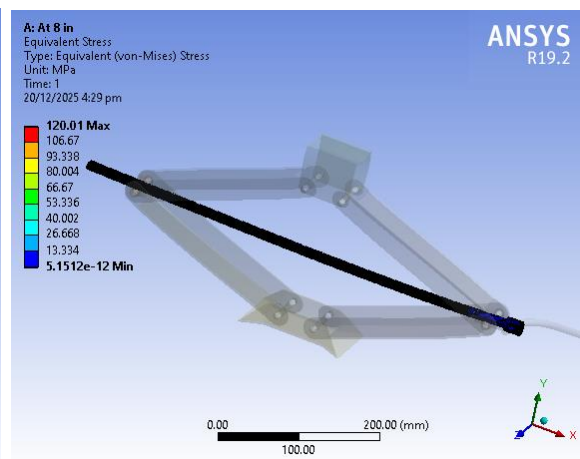
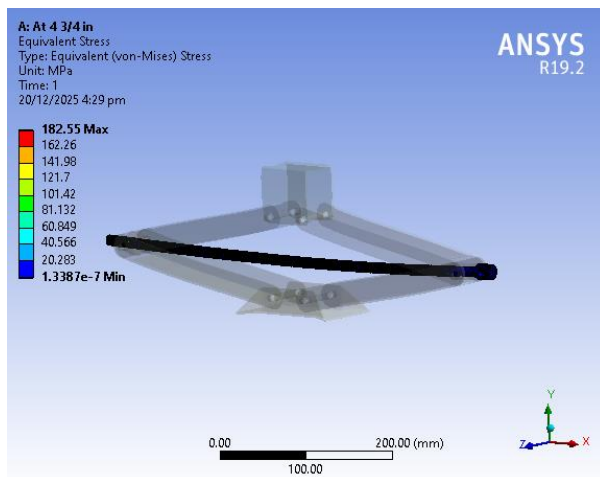
1. Import the IGES geometry into ANSYS Workbench.
2. Define material properties for all components.
3. Generate and refine the finite element mesh.
4. Apply boundary conditions and external loads.
5. Solve the static structural problem for each load and height configuration.
6. Post-process results to obtain von Mises stress, deformation, and factor of safety.
7. Compare results across different heights and loads.

7. Results and Discussion

Von Mises Stress Distribution

The von Mises stress contours indicate that maximum stresses occur near the screw shaft and saddle connection region. As the applied load increases, stress values increase approximately linearly. For a given load, higher saddle heights result in lower stress values due to reduced constraint-induced stiffness and load redistribution along the screw length

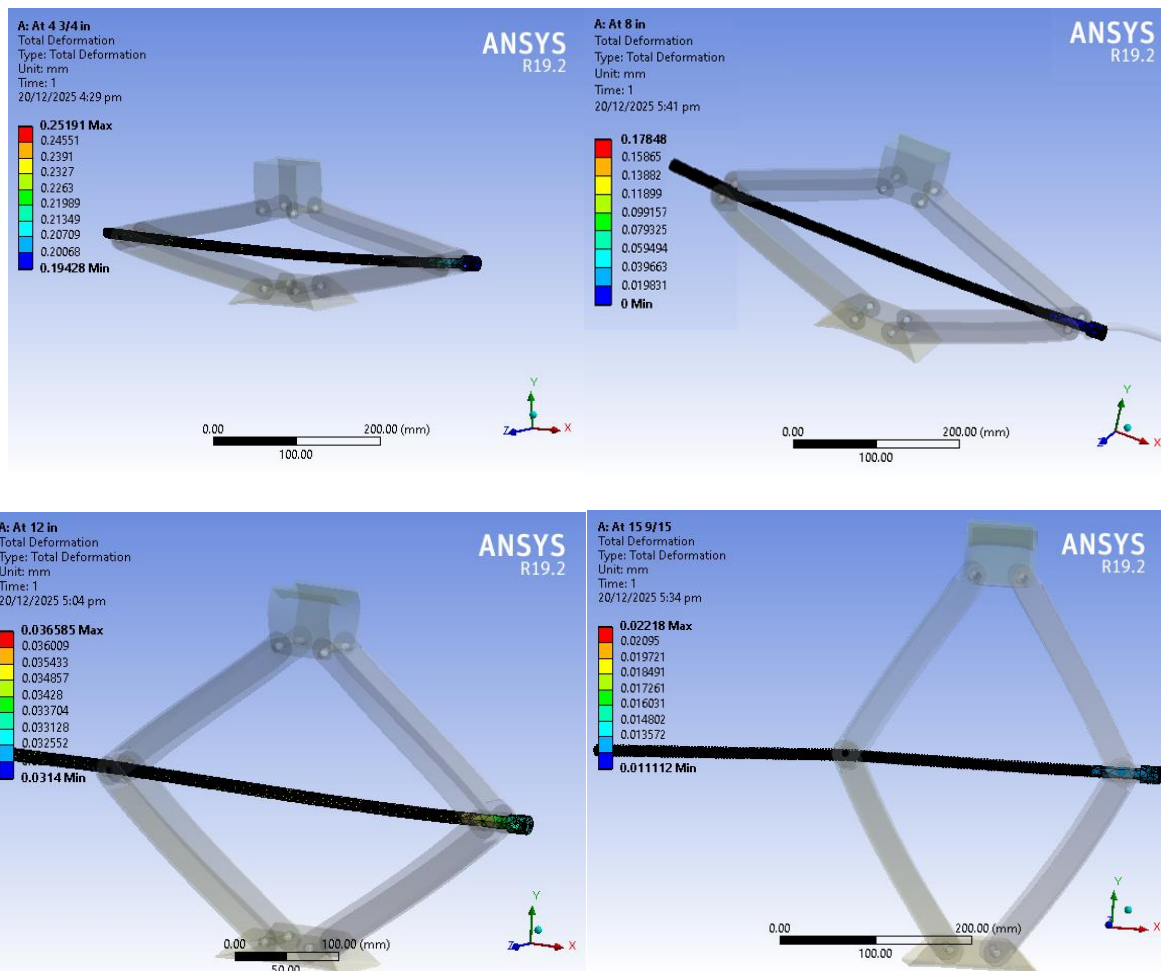
Height (inches)	Max Stress (MPa)	Min Stress (MPa)
4 3/4 in	182.55	1.338×10^{-7}
8 in	120.01	5.1512×10^{-12}
12 in	79.56	1.4978×10^{-12}
15 9/15 in	42.825	1.0036×10^{-12}



Deformation (Y-Direction)

The Y-direction deformation increases with applied load for all configurations. At lower saddle heights, deformation is relatively higher due to larger effective load transfer and reduced structural flexibility. As the saddle height increases, the deformation trend shows a gradual reduction, indicating a more uniform stress distribution along the screw.

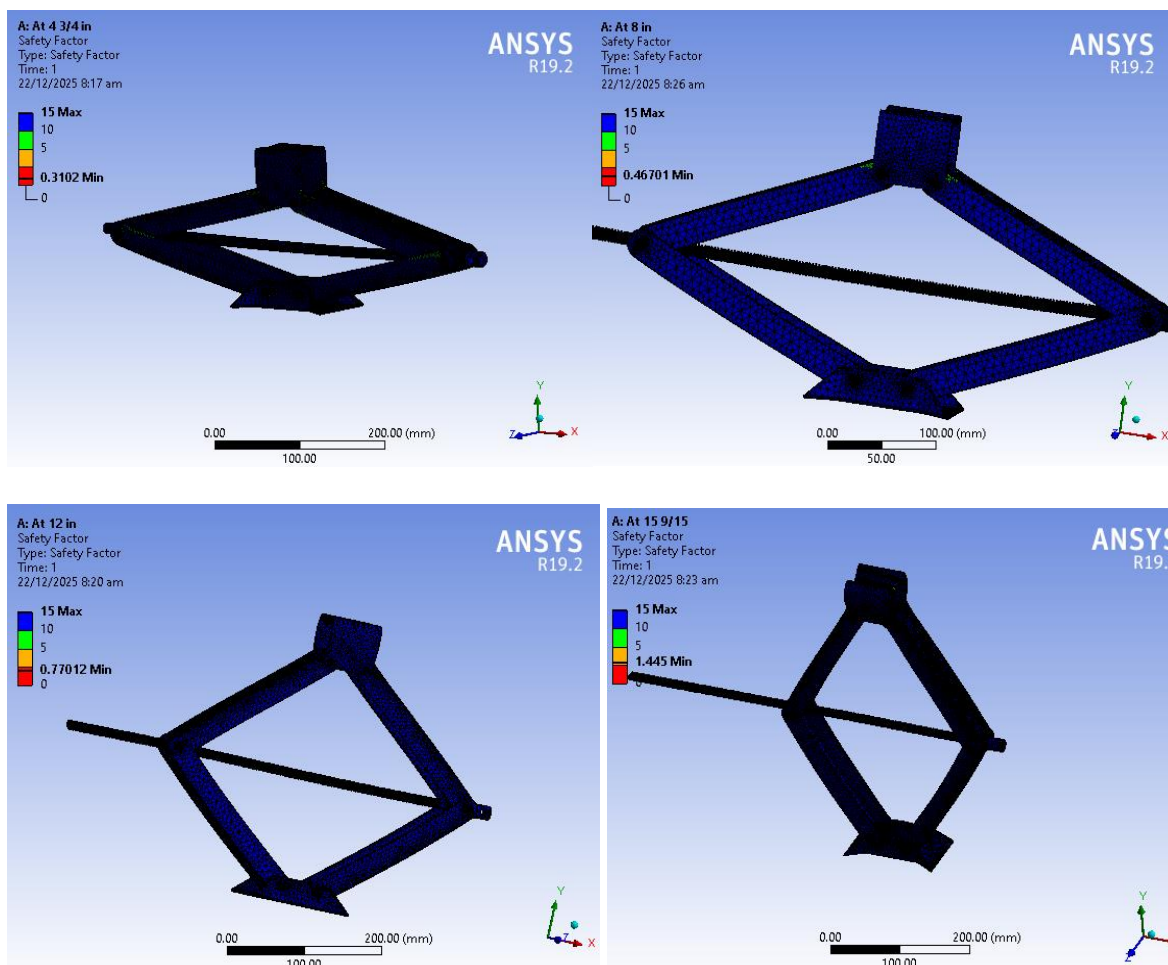
Height (inches)	Max Deformation (mm)	Min Deformation (mm)
4 3/4 in	0.25191	0.19428
8 in	0.17848	0.0
12 in	0.036585	0.0314
15 9/15 in	0.02218	0.011112



Factor of Safety

The factor of safety (FOS) was calculated based on the assumed yield strength of the material. The FOS decreases with increasing load, as expected. However, for all saddle heights and loading conditions considered, the factor of safety remained within acceptable limits, confirming the safe operation of the screw jack under static loading.

Height (inches)	Max Safety Factor	Min Safety Factor
4 3/4 in	15.0	0.3102
8 in	15.0	0.46701
12 in	15.0	0.77012
15 9/15 in	15.0	1.445



8. Comparative Analysis

A comparative study across different saddle heights reveals the following trends:

- Lower saddle heights experience higher stress and deformation.
- Increasing saddle height leads to a reduction in maximum von Mises stress.
- The structural response remains linear within the applied load range.
- The screw jack demonstrates adequate safety margins for all analyzed configurations.

This comparison highlights the importance of considering lifting height during design and safety assessment of screw jacks it is shown in the below tables.

At 4 ³/₄ saddle hieght:

Name ▼	P1 - Force Y Component ▼	P2 - Equivalent Stress Maximum ▼
Units	N ▼	MPa
DP 0 (Current)	-1000	182.55
DP 1	-2000	365.09
DP 2	-3000	547.64
DP 3	-4000	730.19
DP 4	-5000	912.74
DP 5	-6000	1095.3

At 8 in saddle hieght:

Name ▼	P1 - Force Y Component ▼	P1 - Equivalent Stress Maximum ▼
Units	N ▼	MPa
DP 0 (Current)	-1000	120.01
DP 1	-2000	240.01
DP 2	-3000	360.02
DP 3	-4000	480.03
DP 4	-5000	600.03
DP 5	-6000	960.05

At 12 in saddle height:

Name ▼	P1 - Force Y Component ▼	P2 - Equivalent Stress Maximum ▼
Units	N ▼	MPa
DP 0 (Current)	-1000	79.56
DP 1	-2000	159.12
DP 2	-3000	238.68
DP 3	-4000	318.24
DP 4	-5000	397.8
DP 5	-6000	477.36

At 15 7/5 saddle height:

Name ▼	P1 - Force Y Component ▼	P2 - Equivalent Stress Maximum ▼
Units	N ▼	MPa
DP 0 (Current)	-1000	42.825
DP 1	-2000	85.651
DP 2	-3000	128.48
DP 3	-4000	171.3
DP 4	-5000	214.13
DP 5	-6000	256.95

9. Conclusions

The results indicate that the von Mises stress and deformation in the lead screw decrease as the saddle height increases. At lower saddle heights, the effective unsupported length of the lead screw is greater, resulting in higher strain, increased deformation in the Y-direction, and consequently higher stress levels. As the screw jack is extended, the effective length of the lead screw between the supporting arms reduces, leading to lower deformation and reduced stress.

The factor of safety shows an opposite trend with saddle height. The minimum factor of safety is observed at lower saddle heights where stresses are highest, while the factor of safety increases as the saddle height increases due to the reduction in stress. These results confirm that saddle height significantly influences the structural response and safety of the screw jack under static loading conditions.